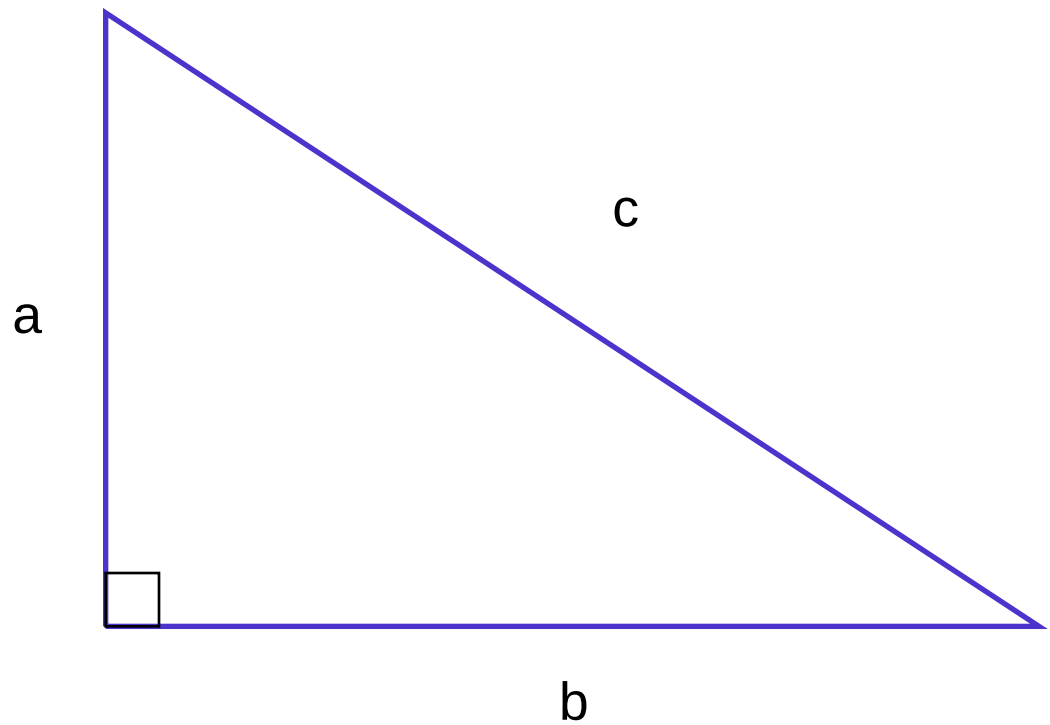
Euclidean geometry

For a right triangle, a^2 + b^2 = c^2.

The right-angle assumption is necessary.

Here c is the hypotenuse, opposite the right angle.